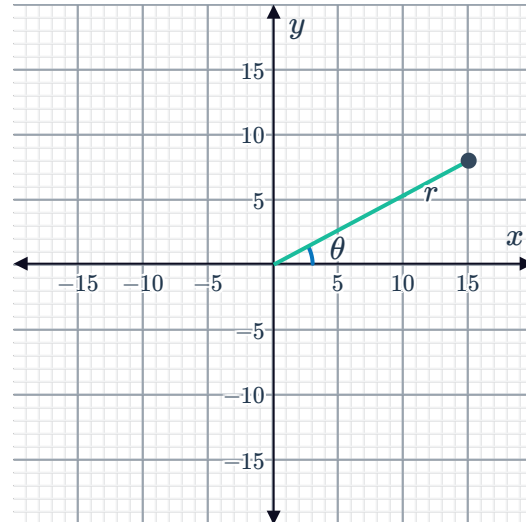


Trigonometric ratios from points



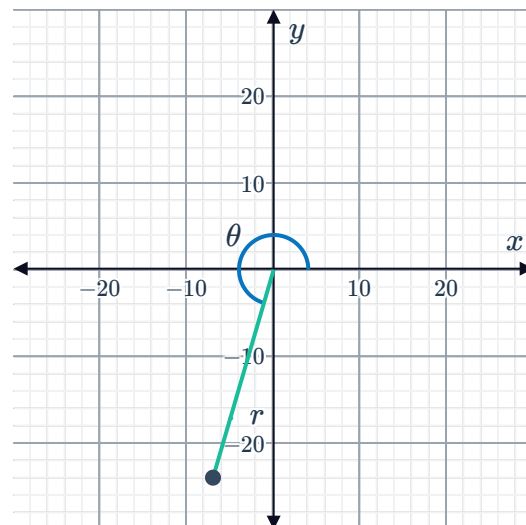
- 1 The point on the following graph has coordinates $(15, 8)$.

- a Find r , the distance from the point to the origin.
- b Find the value of the following:
- | | |
|-------------------|------------------|
| i $\sin \theta$ | ii $\cos \theta$ |
| iii $\tan \theta$ | iv $\csc \theta$ |
| v $\sec \theta$ | vi $\cot \theta$ |



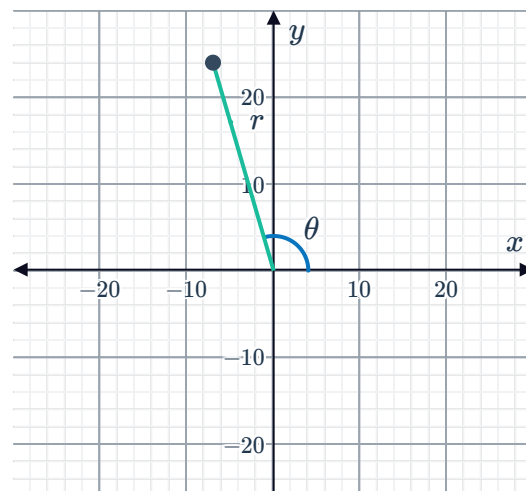
- 2 The point on the following graph has coordinates $(-7, -24)$.

- a Find r , the distance from the point to the origin.
- b Find the value of the following:
- | | |
|-------------------|------------------|
| i $\sin \theta$ | ii $\cos \theta$ |
| iii $\tan \theta$ | iv $\csc \theta$ |
| v $\sec \theta$ | vi $\cot \theta$ |



- 3 The point on the following graph has coordinates $(-7, 24)$.

- a Find r , the distance from the point to the origin.
- b Find the value of the following:
- | | |
|-------------------|------------------|
| i $\sin \theta$ | ii $\cos \theta$ |
| iii $\tan \theta$ | iv $\csc \theta$ |
| v $\sec \theta$ | vi $\cot \theta$ |





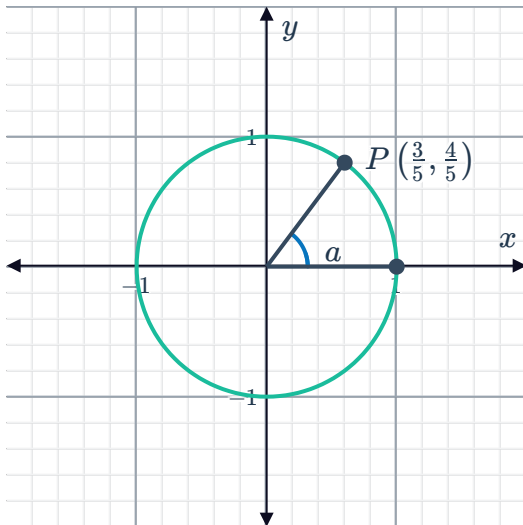
4 For each of the following graphs, find:

i $\sin a$

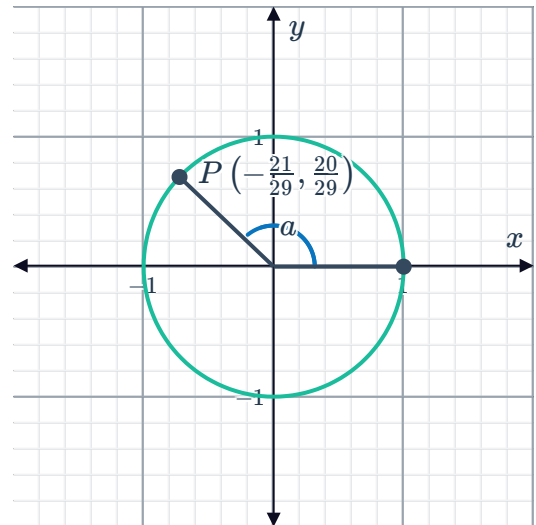
ii $\cos a$

iii $\tan a$

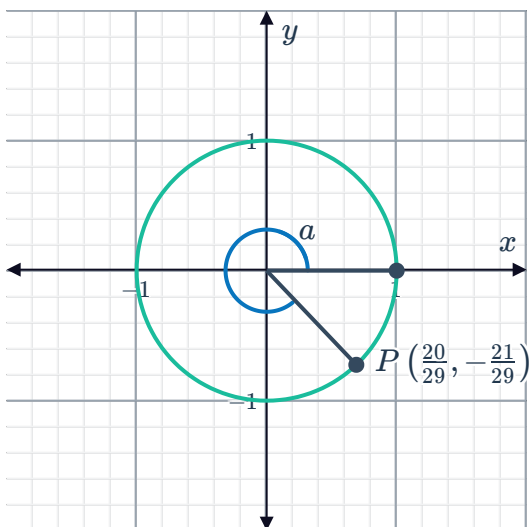
a



b



c



Identify quadrants

5 If $\cos \theta = \frac{-7}{8}$ and θ is in quadrant 3, find $\sin \theta$.



6 State the quadrants in which θ could be located where θ is an angle of rotation corresponding to the point (x, y) on the unit circle such that:

a $\sin \theta > 0$ and $\cos \theta < 0$

b $\sin \theta < 0$ and $\cos \theta < 0$

c $\tan \theta < 0$ and $\cos \theta < 0$

d $\tan \theta > 0$ and $\sin \theta > 0$

e $\sin \theta < 0$ and $\csc \theta < 0$

f $\cos \theta > 0$ and $\sec \theta > 0$

g $\tan \theta > 0$ and $\cot \theta > 0$

h $\cos \theta < 0$ and $\sec \theta > 0$

i $\sec \theta > 0$ and $\csc \theta > 0$

j $\sec \theta > 0$ and $\csc \theta < 0$

k $\cot \theta > 0$ and $\csc \theta > 0$

l $\cot \theta > 0$ and $\sec \theta < 0$

7 State the quadrant where the point $P(x, y)$ is located if P is on the unit circle at a rotation of θ around the origin such that:

a $\sin \theta = \frac{5}{13}$ and $\cos \theta = \frac{12}{13}$

b $\sin \theta = -\frac{12}{37}$ and $\cos \theta = \frac{35}{37}$

c $\sin \theta = -\frac{20}{29}$ and $\cos \theta = -\frac{21}{29}$

d $\sin \theta = \frac{3}{5}$ and $\cos \theta = -\frac{4}{5}$

Equation of the terminal side of an angle

8 The equation of the terminal side of an angle θ in standard position is given by $x + 3y = 0$, for $x \geq 0$.

a Sketch the graph of the equation of the terminal side of θ .

b Using the point $(3, -1)$, find the exact side lengths of the right triangle formed by the terminal side of θ that was graphed in part (a).

c Find the exact values of the following:

i $\sin \theta$

ii $\cos \theta$

iii $\tan \theta$

iv $\csc \theta$

v $\sec \theta$

vi $\cot \theta$



- 9 The equation of the terminal side of an angle θ in standard position is given by $-3x + 2y = 0$, for $x \geq 0$.
- Sketch the graph of the equation of the terminal side of θ .
 - Using the point $(2, 3)$, find the exact side lengths of the right triangle formed by the terminal side of θ that was graphed in part (a).
 - Find the exact values of the following:
 - $\sin \theta$
 - $\cos \theta$
 - $\tan \theta$
 - $\csc \theta$
 - $\sec \theta$
 - $\cot \theta$
- 10 The equation of the terminal side of an angle θ in standard position is given by $4x + y = 0$, for $x \leq 0$.
- Sketch the graph of the equation of the terminal side of θ .
 - Using the point $(-1, 4)$, find the exact side lengths of the right triangle formed by the terminal side of θ that was graphed in part (a).
 - Find the exact values of the following:
 - $\sin \theta$
 - $\cos \theta$
 - $\tan \theta$
 - $\csc \theta$
 - $\sec \theta$
 - $\cot \theta$
- 11 The equation of the terminal side of an angle θ in standard position is given by $x - 4y = 0$, for $x \leq 0$.
- Sketch the graph of the equation of the terminal side of θ .
 - Using the point $(-4, -1)$, find the exact side lengths of the right triangle formed by the terminal side of θ that was graphed in part (a).
 - Find the exact values of the following:
 - $\sin \theta$
 - $\cos \theta$
 - $\tan \theta$
 - $\csc \theta$
 - $\sec \theta$
 - $\cot \theta$